Apple Health + Student Burnout Data

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Overview

I've been tracking my health data through Apple Watch for over six years now — steps, heart rate, exercise minutes, all of it. What started as a casual habit turned into something I kept coming back to, especially around exam weeks when I could literally see my step count drop and my resting heart rate climb. That pattern got me thinking: is there a measurable link between how active I am and how stressed or burned out I feel?

This project is my attempt to answer that question with actual data. I'll be working with a large public dataset on student mental health and burnout (about 1 million records), and then I'll bring in my own Apple Health export to see if the trends I find in the general data hold up in my personal case too.

Data Sources

Student Mental Health & Burnout Dataset (1M rows)

The primary dataset I'm using is a publicly available collection of 1,000,000 synthetic student records. Each record has 20 features — things like study hours, exam pressure, sleep, physical activity, screen time, social support, financial stress, and several mental health indicators (stress level, anxiety score, depression score, burnout score, mental health index, risk level, dropout risk). It's synthetic, which means I need to be careful about how far I generalize from it, but the size makes it useful for finding broad patterns.

My Apple Health Data

This is the part I'm most excited about. I've had an Apple Watch since December 2019, and the export file covers about 874,000 individual records up to March 2026. The metrics I'll focus on are daily step count, heart rate (resting and active), heart rate variability (HRV), active energy burned, exercise minutes, and walking distance. I already know from glancing at the data that my activity dips hard during finals — I want to quantify that properly and see how it compares to what the big dataset suggests about the activity-burnout relationship.

Planned Analysis

I'm splitting this into three parts.The first part is exploratory. I want to dig into the 1M dataset and map out how physical activity, sleep, screen time, and the other lifestyle variables relate to burnout and mental health risk. Which factors matter most? Are there interesting interaction effects — like, does physical activity help less if someone

is also sleeping poorly? Second, I'll run hypothesis tests on my own data. My university's exam schedule gives me natural "treatment" and "control" periods. I want to test whether my step count, HRV, and resting heart rate shift significantly between exam and non-exam weeks. I suspect they do, but I haven't confirmed it statistically yet.

Third, I'll train a few ML models (probably random forest and logistic regression to start) on the large dataset and then feed in features I derive from my Apple Health data. The goal is to see where my own metrics would place me on the mental health risk spectrum the dataset defines. It's a bit of an unusual application since I'm crossing from synthetic population data to real individual data, so I'll need to discuss the limitations of that honestly.

Expected Findings

Based on what I've already seen in my raw data, I'm fairly confident that my physical activity drops and resting heart rate goes up during exam periods. If the large dataset confirms that reduced activity correlates with higher burnout scores at the population level, that would line up nicely.